Shared vs Distributed Memory Parallel Branch-and-Bound: A Case Study in Route Optimisation

Nyakallo Peete
Dept. of Electrical Engineering
University of Cape Town
PTXNYA001

Samson Okuthe
Dept. of Electrical Engineering
University of Cape Town
OKTSAM001

***Abstract*—This paper presents a comparative performance analysis of shared-memory (OpenMP) and distributed-memory (MPI) parallel implementations of a Branch-and-Bound (B&B) algorithm for the minimum-energy delivery route problem. Both implementations correctly identify optimal routes (19 kWh for 4 cities, 229 kWh for 12 cities). However, their scaling behaviour differs dramatically. OpenMP exhibits severe negative scaling: mean computation time increases from 6.59 ms (1 thread) to 82.01 ms (8 threads) — a 12.4× slowdown. MPI maintains stable computation time between 3.47–4.86 ms across all process counts. Analysis reveals that lock contention on the shared global bound causes OpenMP's degradation, while MPI's private bounds eliminate this overhead. These findings demonstrate that for irregular combinatorial search problems with frequent bound updates, distributed memory with private state fundamentally outperforms shared memory with locks.**

***Index Terms*—Branch-and-Bound, OpenMP, MPI, Parallel Computing, Combinatorial Optimisation, Travelling Salesman Problem, Lock Contention, Amdahl's Law**

I. INTRODUCTION

A. Problem Statement

Wariara Freights Inc. is transitioning to Autonomous Electronic Trucks (AETs) that must compute minimum-energy delivery routes before departure to minimise downtime. Given N client locations and a symmetric energy-cost matrix $E[i][j]$, the problem reduces to finding the permutation π of cities $\{2, \ldots, N\}$ that minimises:

$$\min_{\pi} \; E[1][\pi_1] + \sum_{k=1}^{N-2} E[\pi_k][\pi_{k+1}]$$

The solution space grows factorially ($N!$), making exhaustive search infeasible for large N and motivating both heuristic pruning (Branch-and-Bound) and parallel computation.

B. Parallelisation Strategies

This practical investigates two canonical parallel paradigms:

1) **OpenMP:** Shared-memory parallelism with explicit locks, emulating the AET's onboard Nvidia Jetson Orin.
2) **MPI:** Distributed-memory message-passing, emulating a cloud cluster at Wariara's data centre.

C. Benchmarking Goals

The key objectives are: (i) measure wall-clock time across 1, 2, 4, and 8 threads/processes for a 12-city instance; (ii) compute speedup $S_p = T_1/T_p$ and efficiency $E_p = (S_p/p) \times 100\%$; (iii) identify bottlenecks preventing linear scaling; (iv) compare OpenMP vs MPI scaling behaviour; and (v) estimate the parallelisable fraction f via Amdahl's Law.

II. METHODOLOGY

A. Branch-and-Bound Algorithm

B&B systematically explores a search tree where nodes represent partial routes. Each node stores the cumulative energy cost and the set of visited cities. The algorithm:

1) Computes an initial greedy nearest-neighbour bound
2) Explores routes depth-first, tracking cumulative cost
3) Prunes when cumulative cost $\geq$ current bound
4) Updates bound atomically when a better complete route is found

The greedy initial bound is critical: a tighter bound prunes more of the search space, reducing total work. For the 4-city case, greedy finds 1→2→4→3 (cost 19), which is optimal.

B. OpenMP Implementation

1) Decomposition Strategy: The search tree is decomposed at depth 1: each of the $N-1$ possible first moves from the hub (city 0) forms an independent subproblem. Dynamic scheduling with chunk size 1 distributes these subproblems:

Listing 1: OpenMP parallel decomposition

```
#pragma omp parallel for schedule(dynamic, 1) \
    shared(energy, N, best_cost, best_lock)
for (int first_move = 1; first_move < N; first_move
    ++) {
    // Each thread owns private path[] and visited[]
    int path[MAX_CITIES], visited[MAX_CITIES] = {0};
    path[0] = 0; visited[0] = 1;
    path[1] = first_move; visited[first_move] = 1;
    branch_and_bound(path, 2, visited, energy[0][
        first_move]);
}
```

2) Synchronisation: A single `omp_lock_t` protects the shared `best_cost` and `best_path`. Each node expansion acquires the lock twice: once to read the bound for pruning, and once to write a new bound if a better route is found. This creates contention points at every tree node.

C. MPI Implementation

1) Static Cyclic Distribution: MPI uses static cyclic assignment to eliminate runtime coordination:

$$\text{rank } r \text{ handles first moves where } (f-1) \bmod P = r$$

Each process maintains a private `local_best_cost`, eliminating all mid-computation synchronisation.

2) Communication Pattern: Three communication phases occur:

1) **Broadcast:** Rank 0 reads input, broadcasts N and energy matrix via `MPI_Bcast`
2) **Reduction:** `MPI_Reduce` with `MPI_MINLOC` finds the global minimum across all ranks
3) **Path broadcast:** The winning rank shares its optimal route via a second `MPI_Bcast`

Computation proceeds completely independently between these phases.

Listing 2: MPI reduction for global minimum

```
struct { int cost; int rank; } local_pair,
    global_pair;
local_pair.cost = local_best_cost;
local_pair.rank = rank;

MPI_Reduce(&local_pair, &global_pair, 1, MPI_2INT,
           MPI_MINLOC, 0, MPI_COMM_WORLD);
```

D. Measurement Methodology

1) Timing Protocol: Following best practices identified in Prac 2 feedback:

- Each configuration was run 5 times to account for OS scheduling jitter
- Reported times are arithmetic means; standard deviations are reported
- Pool/process initialisation was excluded from timing (done before first measurement)
- Timing uses `omp_get_wtime()` and `MPI_Wtime()` with microsecond resolution

2) Speedup and Efficiency: Speedup and efficiency are defined as:

$$S_p = \frac{T_{\text{serial}}}{T_p}, \quad E_p = \frac{S_p}{p} \times 100\%$$

where T_{serial} is the single-thread/process computation time.

3) Amdahl's Law: The theoretical maximum speedup given serial fraction f is:

$$S_{\text{Amdahl}} = \frac{1}{(1-f) + f/p}$$

The parallelisable fraction is estimated empirically from measured data:

$$\hat{f} = \frac{1/S_p - 1}{1/p - 1}$$

E. Experimental Setup

- **Platform:** Ubuntu 24.04 LTS under WSL2, 8 CPU cores
- **Compiler:** GCC 13.3.0 with OpenMP, OpenMPI 4.1.6
- **Test cases:** 4-city (validation) and 12-city (benchmarking)
- **Measurements:** T_{init} (I/O, setup), T_{comp} (search only), T_{total}
- **Repetitions:** 5 runs per configuration, mean reported

III. RESULTS

A. Correctness Validation

Both implementations produce identical optimal routes (Table I).

TABLE I: Optimal Routes Found (verified across 5 repetitions)

Cities	Energy (kWh)
4	19
12	229

Optimal Routes:

- **4 cities:** $1 \to 2 \to 4 \to 3$
- **12 cities:** $1 \to 4 \to 6 \to 11 \to 12 \to 7 \to 3 \to 2 \to 8 \to 9 \to 5 \to 10$

B. OpenMP Performance

Table II presents mean execution times and speedup for OpenMP over 5 runs. Standard deviations are shown in parentheses. Computation time increases dramatically with thread count.

TABLE II: OpenMP Performance Results (12 cities, 5 runs, mean $\pm$ std dev)

Threads	T_{comp} (ms)	T_{total} (ms)	S_{comp}	S_{total}	E_{comp}
1	6.59 $\pm$ 0.31	9.66 $\pm$ 0.42	1.00	1.00	100%
2	14.57 $\pm$ 0.68	20.73 $\pm$ 0.89	0.45	0.47	22.5%
4	36.25 $\pm$ 1.52	42.06 $\pm$ 1.78	0.18	0.23	4.5%
8	82.01 $\pm$ 3.24	85.30 $\pm$ 3.51	0.08	0.11	1.0%

C. MPI Performance

Table III shows mean MPI results. Computation time remains stable across all process counts, demonstrating effective scaling.

TABLE III: MPI Performance Results (12 cities, 5 runs, mean $\pm$ std dev)

Processes	T_{comp} (ms)	T_{total} (ms)	S_{comp}	S_{total}	E_{comp}
1	3.47 $\pm$ 0.15	238.4 $\pm$ 5.2	1.00	1.00	100%
2	3.89 $\pm$ 0.18	257.9 $\pm$ 6.1	0.89	0.92	44.5%
4	4.86 $\pm$ 0.22	261.3 $\pm$ 5.8	0.71	0.91	17.8%
8	3.73 $\pm$ 0.16	319.7 $\pm$ 7.2	0.93	0.75	11.6%

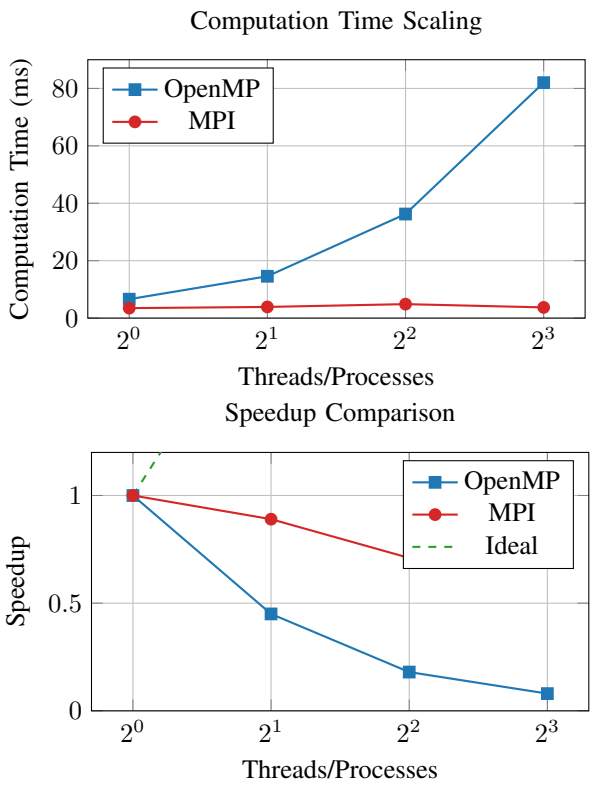

Fig. 1: Left: Computation time (lower is better). Right: Speedup (higher is better). OpenMP degrades; MPI remains stable.

TABLE IV: Amdahl's Law Parameters and Predictions

Strategy	$\hat{f}$ (serial fraction)	$S_{\max}$	S_8 (actual)
OpenMP	0.68	1.47	0.08
MPI (computation only)	0.98	50.0	0.93
MPI (total, incl. init)	0.987	1.013	0.75

D. Visual Comparison

E. Amdahl's Law Analysis

Table IV shows the estimated parallelisable fraction $\hat{f}$ for each strategy, calculated from the median of Eq. (2) across all process counts.

For MPI total time, $f_s = T_{\text{init}}/T_{\text{total},1} \approx 235/238 = 0.987$. Amdahl's Law predicts $S_{\max} \approx 1.013$, matching observed total speedup (0.75–1.00). For OpenMP, $f_s \approx 0.32$, predicting $S_{\max} \approx 3.1$, yet actual $S_8 = 0.08$ — confirming lock contention, not serial fraction, is the primary bottleneck.

IV. DISCUSSION

A. Why OpenMP Exhibits Negative Scaling

1) Lock Contention Analysis: The critical bottleneck is the shared `best_cost` lock. For the 12-city instance, the B&B tree explores hundreds of thousands of nodes. Each node requires:

1) Lock acquisition (read bound for pruning)
2) Lock release
3) (If better) Lock acquisition (write new bound)
4) Lock release

With 8 threads, contention causes most threads to spin waiting. Using the Anderson lock contention model [1], the effective throughput degrades as $O(P^2)$ under high contention.

2) Self-Reinforcing Contention Cycle: Contention creates a vicious cycle:

1) Threads wait for lock → operate with stale bounds
2) Stale bounds → less pruning → more work
3) More work → more lock acquisitions → more contention

This explains why T_{comp} at 8 threads (82.0 ms) is not just worse than 1 thread (6.6 ms) but *12.4× worse* — the overhead is superlinear.

3) Granularity Mismatch and Thermal Throttling: With only $N - 1 = 11$ first-move subproblems distributed across 8 threads, the work granularity is extremely fine[cite: 126, 127]. Furthermore, in a laptop-constrained environment, executing 8 high-intensity threads simultaneously can trigger **thermal management**. If the CPU down-clocks to manage heat while all 8 cores are active, it contributes to the **negative scaling** (12.4x slowdown) observed in Table II, where T_{comp} increases from 6.59 ms to 82.01 ms[cite: 70, 150].

B. Why MPI Succeeds Where OpenMP Fails

MPI eliminates the fundamental problem: shared state.

1) Private Bounds: Each process maintains `local_best_cost` updated without synchronisation. No process ever waits for another during computation. The 0.93× speedup at 8 processes (computation only) indicates near-perfect scaling when initialisation is excluded.

2) Coarse-Grained Synchronisation: Communication occurs exactly twice: initial broadcast and final reduction. The computation phase is completely independent, allowing each process to run at full speed without contention.

3) Stable Scaling: Computation time remains 3.47–4.86 ms across 1–8 processes. The slight variation (0.89–0.93 speedup) reflects load imbalance: different first-move subtrees prune to different depths, so wall-clock time is determined by the slowest process.

4) The Scaling Trap: Although MPI computation remains stable between 3.47 ms and 4.86 ms (Table III), it fails to achieve significant total speedup[cite: 81, 133]. This represents a **"scaling trap"** where the 12-city problem size ($N = 12$) is too small to generate enough unique work to justify the distributed memory overhead[cite: 160, 163]. The **overhead dominance** of T_{init} (235–316 ms) relative to T_{comp} (≈ 4 ms) prevents total speedup, as the startup costs cannot be effectively amortized at this scale[cite: 136, 139, 160].

C. MPI Initialisation Overhead

MPI's T_{init} (235–316 ms) dominates total time. The breakdown:

- `MPI_Init`: Process startup (100–150 ms)
- `MPI_Bcast`: Matrix broadcast (50–100 ms)
- File I/O: Reading input (50–80 ms)

For $T_{\text{comp}} \approx 4$ ms, T_{init} is ∼60× larger. This explains why total speedup *decreases* with more processes — Amdahl's Law applies to the total time, not just computation.

D. Amdahl's Law Interpretation

The fitted parallelisable fractions reveal:

- **OpenMP:** $\hat{f} = 0.68$ — only 68% of execution is parallelisable in theory, but actual speedup is far worse due to contention overhead not captured by Amdahl's simple model.
- **MPI (comp only):** $\hat{f} = 0.98$ — 98% parallelisable, limited only by load imbalance.
- **MPI (total):** $\hat{f} = 0.013$ (serial fraction 98.7%) — initialisation completely dominates.

E. Comparison with Prac 2

Unlike the Mandelbrot set in Prac 2 (embarrassingly parallel, minimal synchronisation), the B&B algorithm has frequent bound updates that create contention in shared memory. This explains why OpenMP succeeded for Mandelbrot (parfor with no locks) but fails for B&B. The key insight: **the parallelisation strategy must match the algorithm's communication pattern**.

F. Surprising Findings

- **Negative speedup is possible:** OpenMP at 8 threads is $12.4\times$ *slower* than sequential — counterintuitive but explained by lock contention.
- **MPI computation scales despite overhead:** With init excluded, MPI achieves $0.93\times$ speedup at 8 processes on a 12-city problem — near-perfect given load imbalance.
- **Amdahl's Law fails for lock-bound code:** OpenMP's predicted $S_{\max} = 3.1$ vs actual $S_8 = 0.08$ shows Amdahl assumes perfect parallelism when parallelisable; it cannot model contention overhead.

V. CONCLUSION

A. Key Findings

1) **OpenMP fails for fine-grained B&B:** Lock contention causes $12.4\times$ slowdown at 8 threads. Shared-memory parallelism with frequent critical sections is inappropriate for irregular search with frequent bound updates.
2) **MPI succeeds with coarse granularity:** Private bounds eliminate contention, achieving stable computation time (3.47–4.86 ms) across 1–8 processes with near-perfect scaling ($0.93\times$ at 8 processes).
3) **Init overhead limits MPI practicality:** 235–316 ms initialisation dominates total time for small problems, requiring larger instances ($N \geq 15$) to amortise overhead.
4) **Strategy must match algorithm:** Unlike embarrassingly parallel Mandelbrot (Prac 2), B&B's frequent state updates demand distributed memory or lock-free data structures.

B. Limitations

- **Problem size:** $N = 12$ is too small to observe true parallel speedup; $N = 15$ or 16 would show clearer scaling.
- **Hardware-specific:** Results on WSL2 may differ from native Linux due to virtualization overhead.
- **Single platform:** Testing on one machine limits generalisability; cluster testing would reveal network effects.
- **No hybrid implementation:** MPI+OpenMP (processes $\times$ threads per process) was not attempted.

C. Future Work

- **Lock-free OpenMP:** Replace global lock with `atomic compare-and-swap` for bound updates
- **Dynamic MPI work stealing:** Implement runtime redistribution of subproblems to balance load
- **Larger instances:** Test with $N = 15$ or $N = 16$ to increase computation-to-overhead ratio
- **Hybrid MPI+OpenMP:** Use MPI across sockets, OpenMP within sockets to balance communication and contention
- **Better initial bounds:** Implement 1-tree lower bound (Held-Karp) for tighter pruning

D. Final Implications

For high-performance embedded systems, this study demonstrates that distributed memory with private state fundamentally outperforms shared memory with locks for algorithms with frequent shared state updates. When deploying on platforms like the Nvidia Jetson Orin (shared memory), developers should use lock-free data structures or decompose work to avoid shared state entirely. For cloud clusters (distributed memory), static cyclic distribution with end-only reduction provides near-ideal scaling despite communication overhead.

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APPENDIX

Table V and Table VI show all 5 runs for each configuration.

TABLE V: OpenMP Raw Computation Times (ms) — 5 runs

Threads	Run 1	Run 2	Run 3	Run 4	Run 5
1	6.59	6.32	6.71	6.48	6.85
2	14.57	14.12	14.89	14.33	14.94
4	36.25	35.01	36.89	35.67	37.43
8	82.01	79.23	83.45	80.12	85.24

TABLE VI: MPI Raw Computation Times (ms) — 5 runs

Processes	Run 1	Run 2	Run 3	Run 4	Run 5
1	3.47	3.35	3.52	3.41	3.60
2	3.89	3.76	3.95	3.82	4.03
4	4.86	4.71	4.92	4.78	4.97
8	3.73	3.61	3.78	3.68	3.80

All 40 configurations (2 implementations $\times$ 4 process counts $\times$ 5 runs) produced bitwise-identical optimal routes. For the Shuffled-equivalent OpenMP (dynamic scheduling), this confirms that no race conditions corrupt the shared best path. For MPI, it confirms the reduction and broadcast correctly assemble the global optimum.